Diffusion Seminar (Week8)

Diffusion Language Models I

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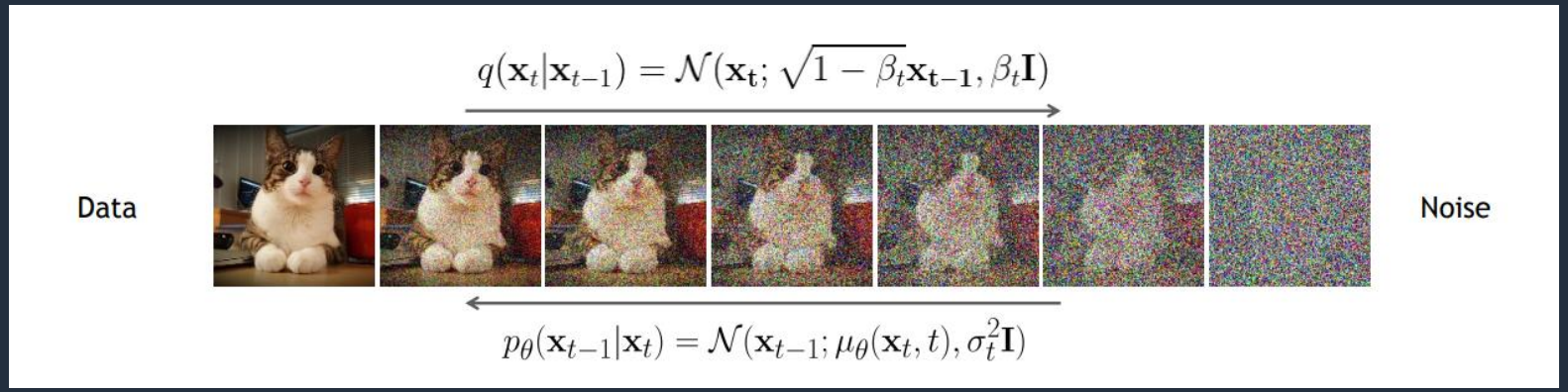
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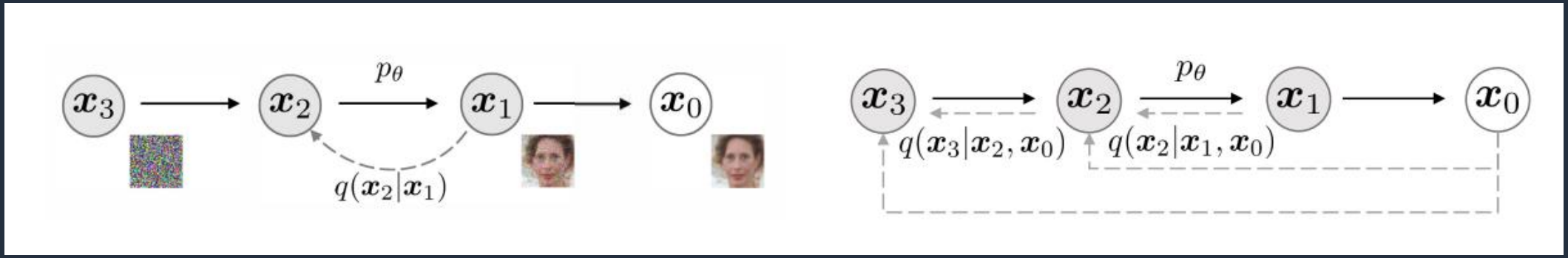
From Vision to Language

Recap: DDPM, DDIM

- **DDPM**: Forward process gradually adds Gaussian noise, backward process learns to denoise step-by-step.



- **DDIM**: Deterministic, non-Markovian sampling, fewer steps, better control over trade-off between speed and quality.



Recap: Diffusion in Vision

- These models rely on continuous corruption and denoising in a fixed-dimensional space.

Latent Diffusion



DiT



R. Rombach et al., "High Resolution Image Synthesis with Latent Diffusion Models", CVPR 2022.

Peebles & Xie., "Scalable Diffusion Models with Transformers", CVPR 2023.

Recap: Diffusion in Vision

- These models rely on continuous corruption and denoising in a fixed-dimensional space.

Classifier Guidance



Classifier-Free Guidance

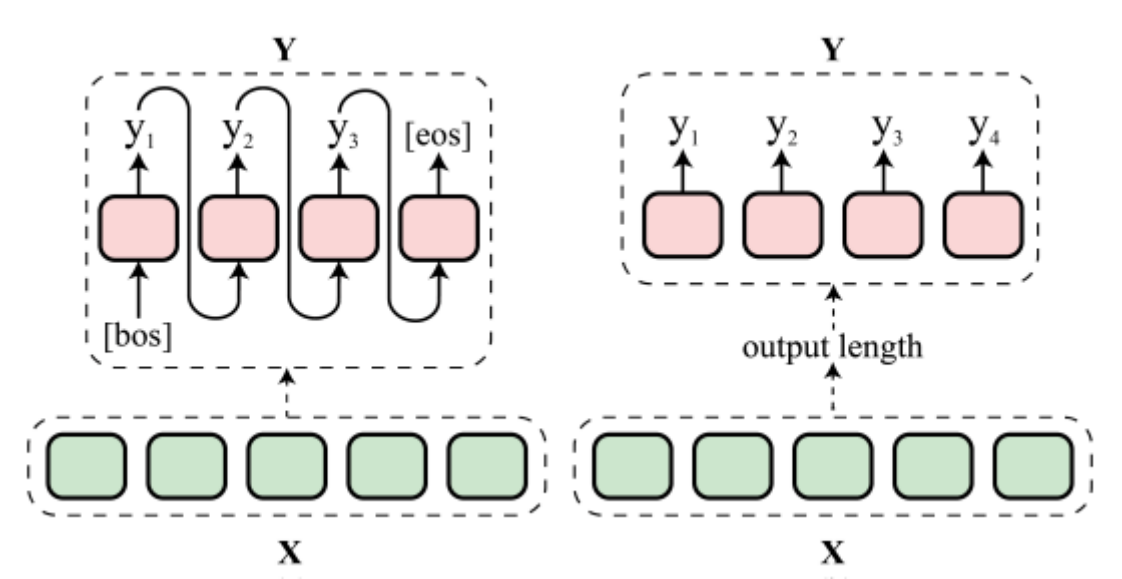


Dhariwal et al. "Diffusion Models Beat GANs on Image Synthesis" , NeurIPS 2021

Ho & Salimans., "CLASSIFIER-FREE DIFFUSION GUIDANCE", NeurIPS 2021.

Bringing Diffusion to Language

- Diffusion models work well in the vision domain.
- Then, can we apply the diffusion process to ***language generation?***

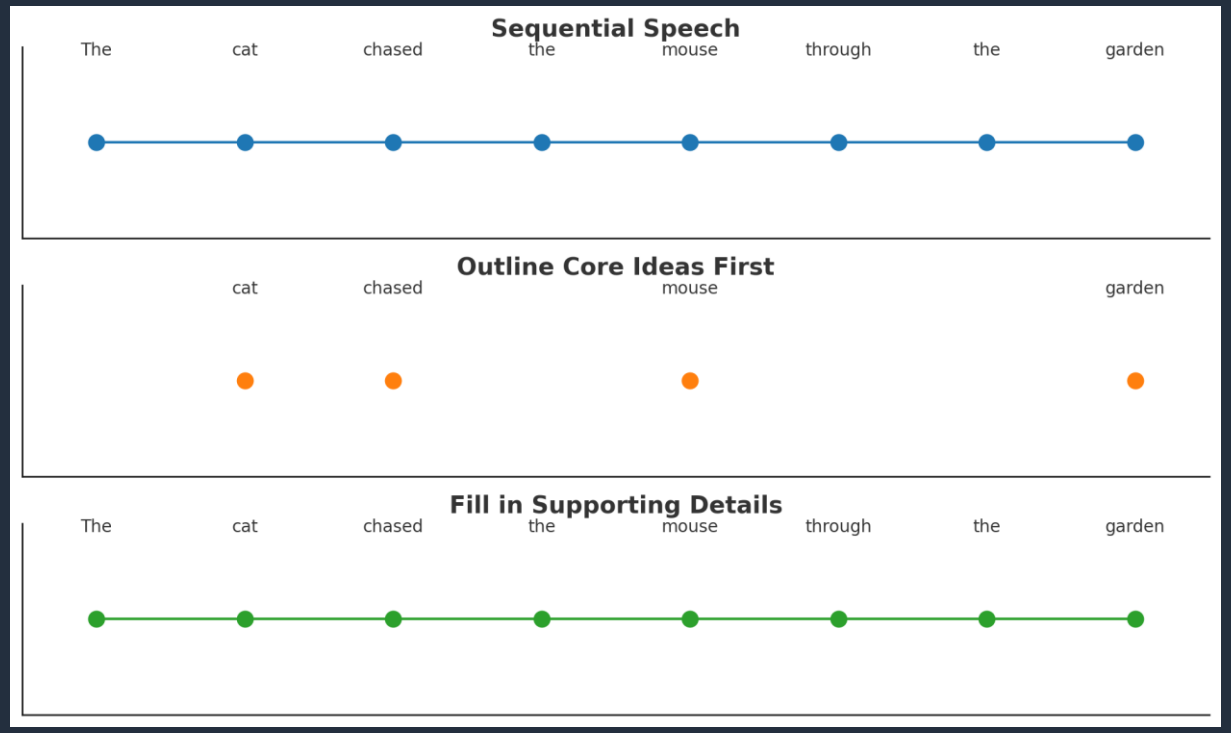


- First, since humans typically speak sequentially, it seems natural to model language generation as a sequential process
- Then, why look beyond sequential generation and explore a parallel, iterative framework like diffusion?

Su et al., "Non-Autoregressive Text Generation with Pre-trained Language Models", EACL 2021.

Simple Rationale for Using Diffusion in Text

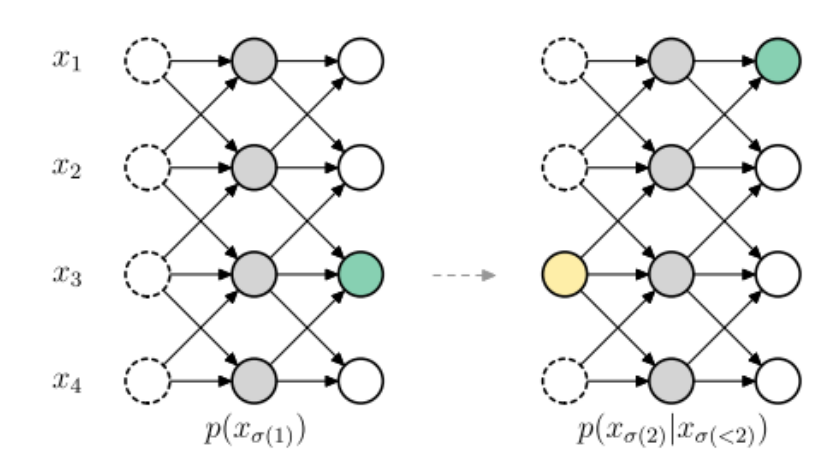
- While humans speak sequentially, we do not always produce a full sequence strictly from start to end.
- We sometimes outline key ideas or core words first, then fill in supporting details and refine the structure.



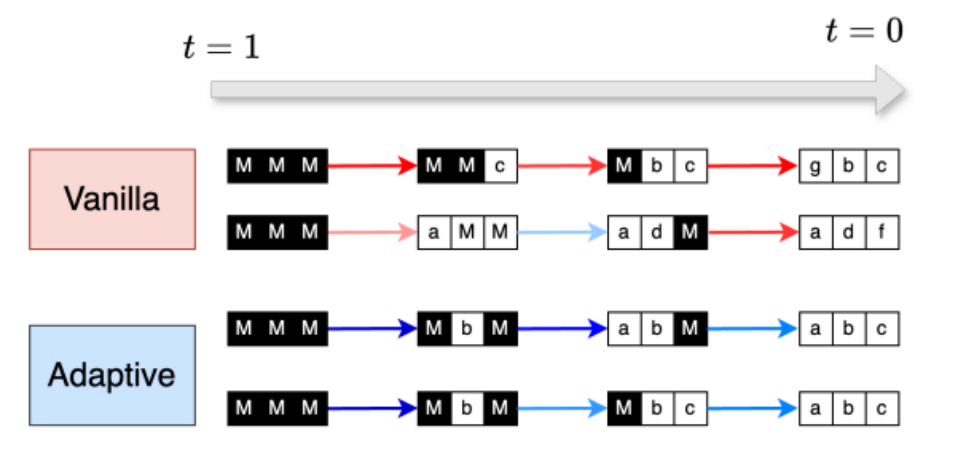
Diffusion as the generalization of AR-based approach

- Recent observations show that left-to-right decoding can be interpreted as a specific, degenerate case of diffusion modeling with a fixed generation order.
- A flexible generation order allows the model to leverage bidirectional context, avoid early-error lock-in, and adapt generation strategy to the task.

Autoregressive Diffusion Models
(Hoogeboom et al., 2022)

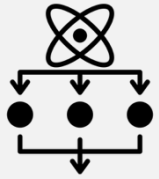


Understanding Token Ordering in Masked Diffusions (Kim et al., 2025)



Why Use Diffusion Language Models?

- Autoregressive (AR) models still perform well.
- So why should we consider using diffusion for language models?



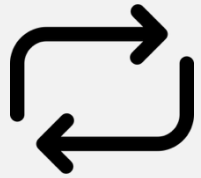
Parallel Decoding

All tokens can be generated simultaneously, utilizing more hardware



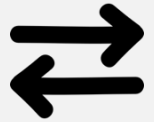
Controllability

Generation could be easily steered with Classifier guidance, Classifier-free guidance



Iterative Refinement

Allows correction of early mistakes and globally coherent outputs

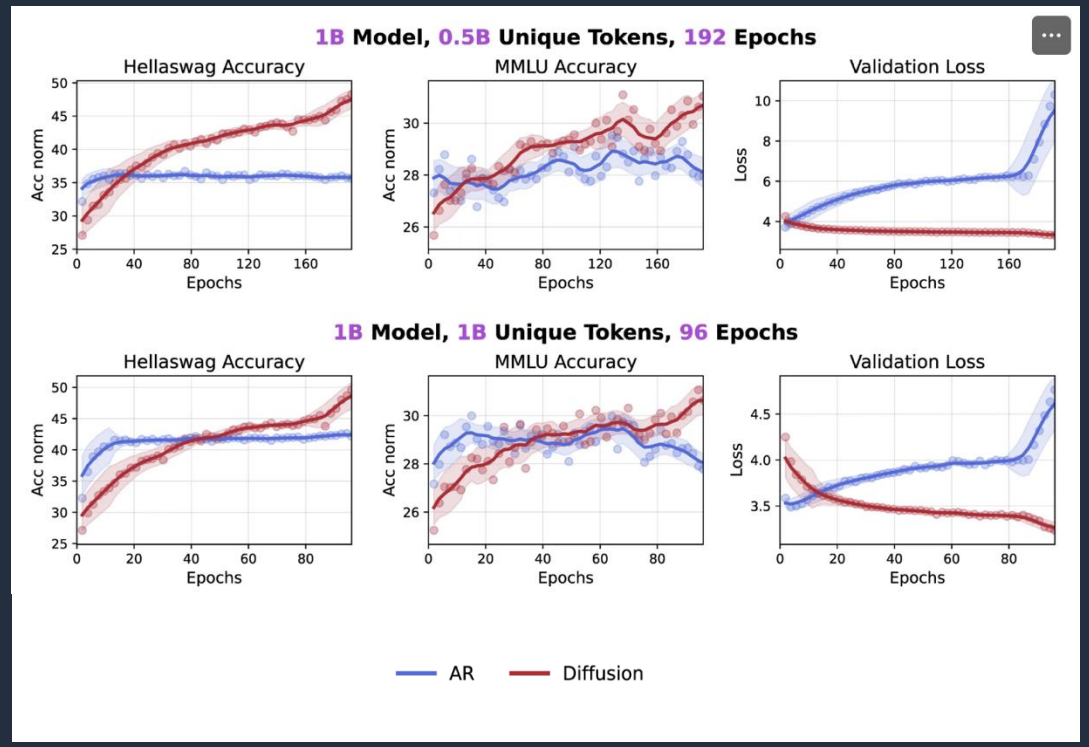


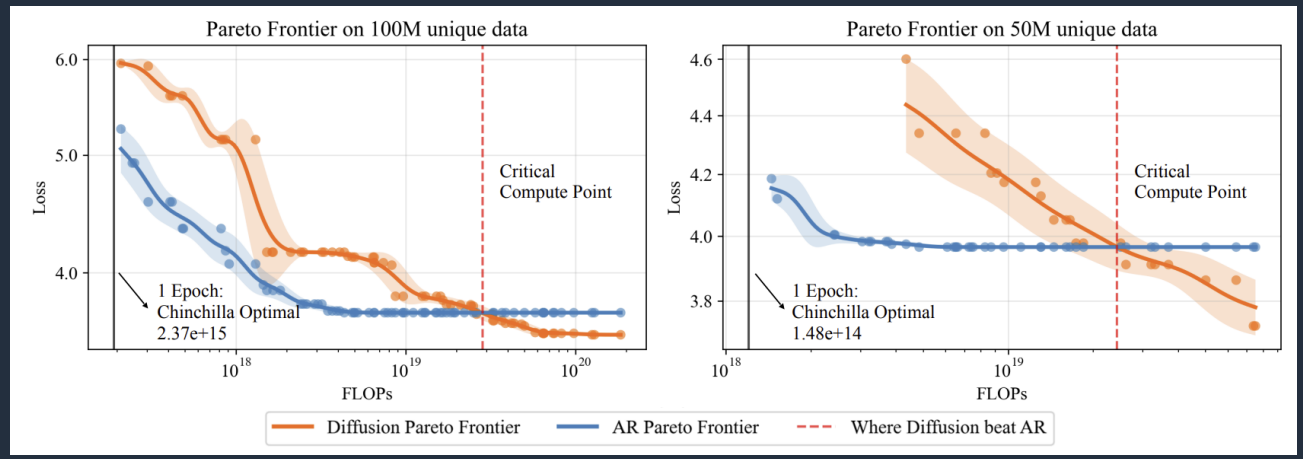
Bidirectional Context

Considers full sequence context during denoising, enabling infilling

Why Use Diffusion Language Models?

- **Better in Data-Constrained Settings**: When data is scarce but compute is abundant, diffusion models can outperform autoregressive (AR) models.





Ni et al., "Diffusion Language Models are Super Data Learners", 2025.

The Challenge: Discreteness of Text



Why Diffusion Doesn’t Directly Work on Text

For image, when noised, still a valid image at each step



Why Diffusion Doesn’t Directly Work on Text

- Unlike images, text is made of discrete tokens
  -> we can’t smoothly add noise like we do with pixels.

For text tokens

Text	Tokens
The cat sits on the mat	0, 40, 43, 13, 2, 67
↓	
? ? ? ? ? ?	0.5, 40.71, 42.1, 13.4, 1.8, 66.4
↓ Rounding?	
No image why on the love	1, 41, 42, 13, 2, 66

Fragility of Semantics

- In images, a small perturbation in a single pixel (e.g., brown → red) is less noticeable and rarely changes the overall meaning.

- In text, even a single token change (e.g., “a” → “b” or “cat” → “car”) can completely alter meaning and grammaticality.

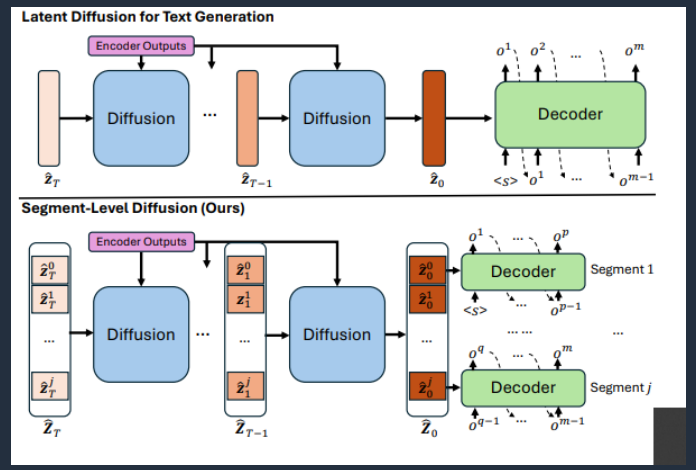
“My cat ate a fish” → “My car ate b fish”

- Thus, it’s harder for diffusion in text to ensure that noisy intermediate states map back to fluent, coherent sentences.

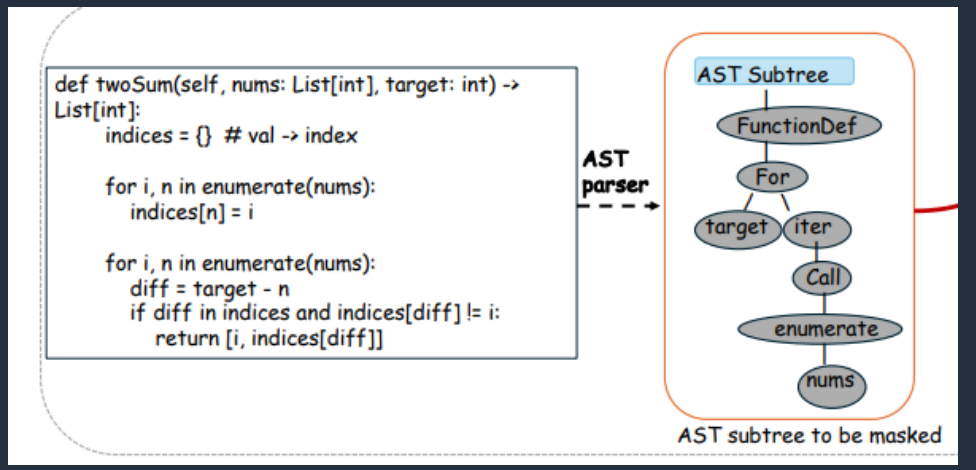
Breaking Token-Level Coherence

- Token-level corruption in diffusion assumes conditional independence between positions at each denoising step.
- This can break sequential dependencies (syntax, semantics, ordering), especially in structured or long-form text.

Segment-Level Diffusion
(Zhu et al., 2025)



TreeDiff
(Ren et al., 2025)



Zhu et al., "Segment-Level Diffusion", ACL, 2025.

Ren et al., "TreeDiff: AST-Guided Code Generation with Diffusion LLMs", 2025.

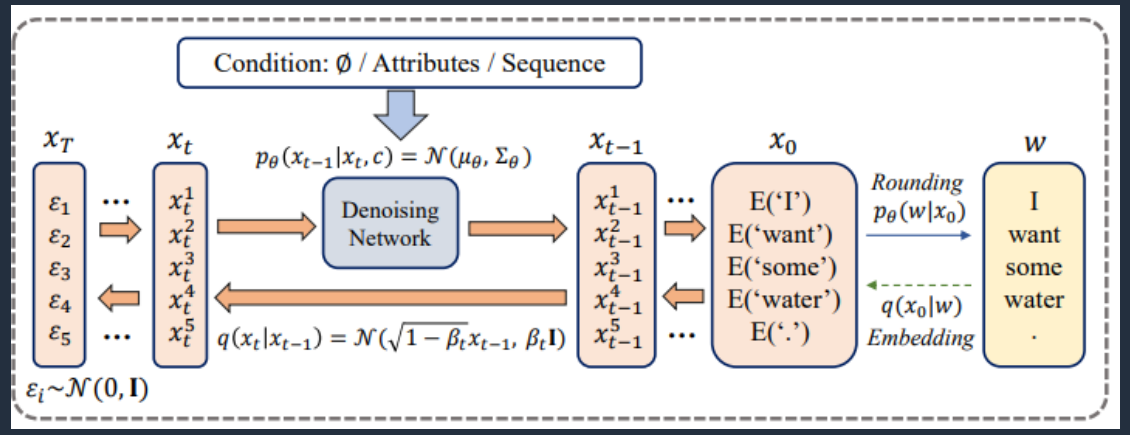
Two Big Families of DLMs

Dealing with Discreteness

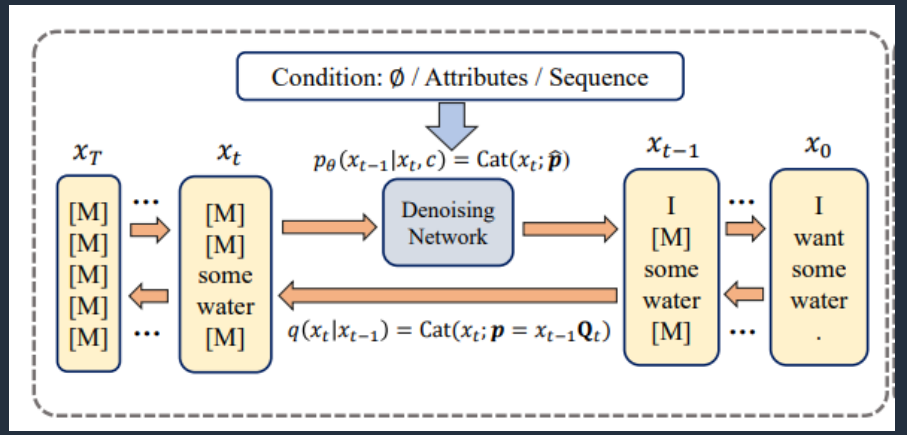
- To apply diffusion to language, we need to bridge the gap between discrete tokens and the continuous noise-based diffusion framework.

Two main paradigms

Continuous Diffusion



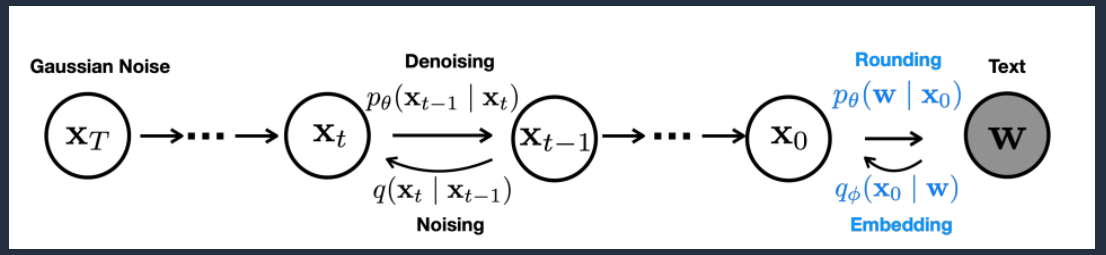
Discrete Diffusion



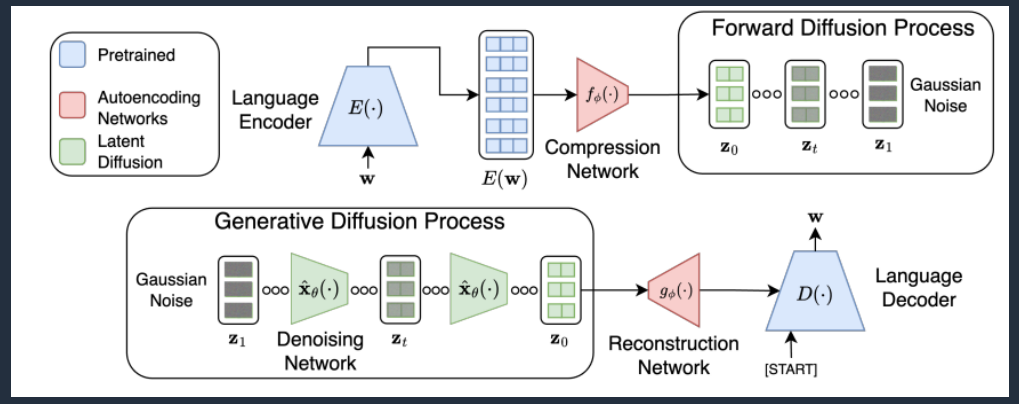
Continuous Diffusion Language Model

- Instead of adding noise directly to tokens, we embed tokens into a continuous space and apply diffusion there.

Embedding space
(Diffusion-LM, 2022)



Latent space
(LD4LG, 2022)



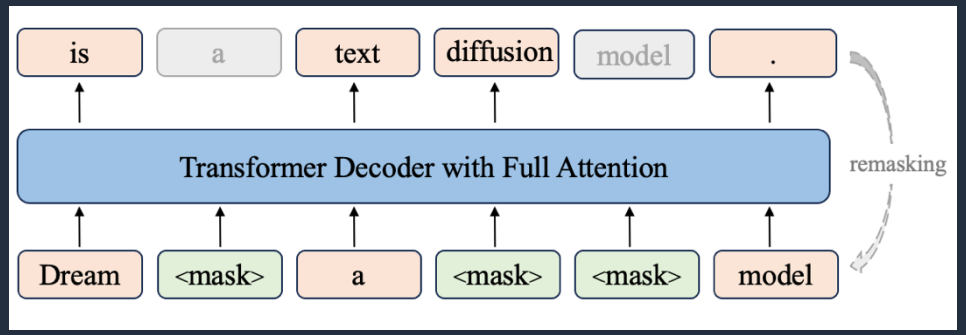
- Diffusion-LM (2022), LD4LG (2022), Genie (2023), AR-Diffusion (2024)...

https://arxiv.org/abs/2205.14217 https://arxiv.org/pdf/2212.09462

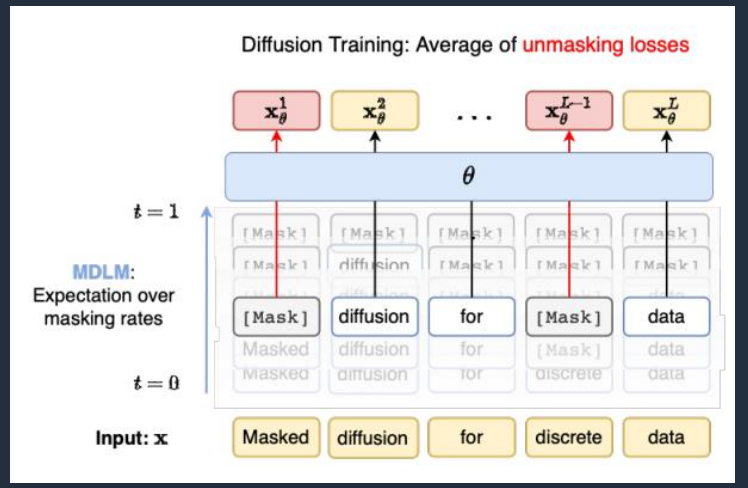
Discrete Diffusion Language Model

- Instead of embedding tokens into a continuous space, discrete diffusion models operate directly in token space by progressively corrupting the input sequence through symbolic operations.
- Random replace, mask or shuffle

Dream
(Ye et al., 2025)



MDLM
(Sahoo et al., 2024)



- DiffusionBERT (2022), MDLM (2024), SEDD (2024), LLADA (2025)...

https://hkunlp.github.io/blog/2025/dream/

https://arxiv.org/pdf/2406.07524

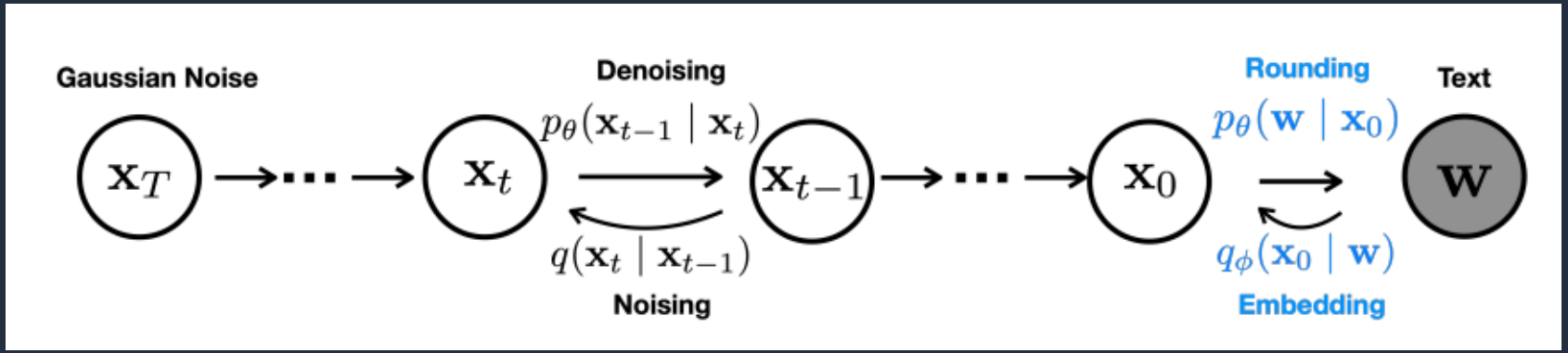
Continuous VS. Discrete Diffusion

	Continuous Diffusion LM	Discrete Diffusion LM
Space	Embedding, Latent	Token
Noising process	Continuous noising	Masking, Replacing
Pros	• Faster single-step operations (low-dim) • Classifier guidance • Image-based techniques are applicable	• Works directly on tokens • Easier alignment with existing models • Simplified training objective
Cons	• Training instability • Scaling to large corpora remains unproven • Requires encoder/decoder modules	• Classifier guidance is limited • Transition itself is not differentiable

Continuous Diffusion Language Models

Diffusion-LM

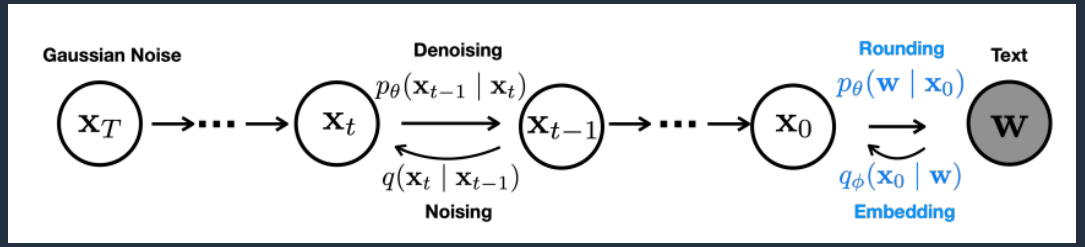
- Diffusion-LM introduces embedding and rounding steps to map discrete tokens into a continuous domain suitable for diffusion.



- Forward pass: $q(x_t|x_{t-1}) = N(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$
- Backward pass: $p_\theta(x_{t-1}|x_t) = N(x_{t-1}; \mu_\theta(x_t, t), \sigma_t{}^2 I)$

} Same as in DDPM

Diffusion-LM



- Objective function

DDPM objective

$$\mathcal{L}_{\text{simple}}(\mathbf{x}_0) = \sum_{t=1}^{T} \mathop{\mathbb{E}}_{q(\mathbf{x}_t|\mathbf{x}_0)} ||\mu_\theta(\mathbf{x}_t, t) - \hat{\mu}(\mathbf{x}_t, \mathbf{x}_0)||^2$$

↓

Embedding & Rounding added

$$\mathcal{L}^{\text{e2e}}_{\text{simple}}(\mathbf{w}) = \mathop{\mathbb{E}}_{q_\phi(\mathbf{x}_{0:T}|\mathbf{w})} \left[\mathcal{L}_{\text{simple}}(\mathbf{x}_0) + ||\text{EMB}(\mathbf{w}) - \mu_\theta(\mathbf{x}_1, 1)||^2 - \log p_\theta(\mathbf{w}|\mathbf{x}_0)\right]$$

↓

Predict x_0 instead

$$\mathcal{L}^{\text{e2e}}_{\mathbf{x}_0\text{-simple}}(\mathbf{x}_0) = \sum_{t=1}^{T} \mathbb{E}_{\mathbf{x}_t} ||f_\theta(\mathbf{x}_t, t) - \mathbf{x}_0||^2$$

Li et al., "Diffusion-LM Improves Controllable Text Generation", NeurIPS, 2022.

Diffusion-LM

- Classifier-guidance

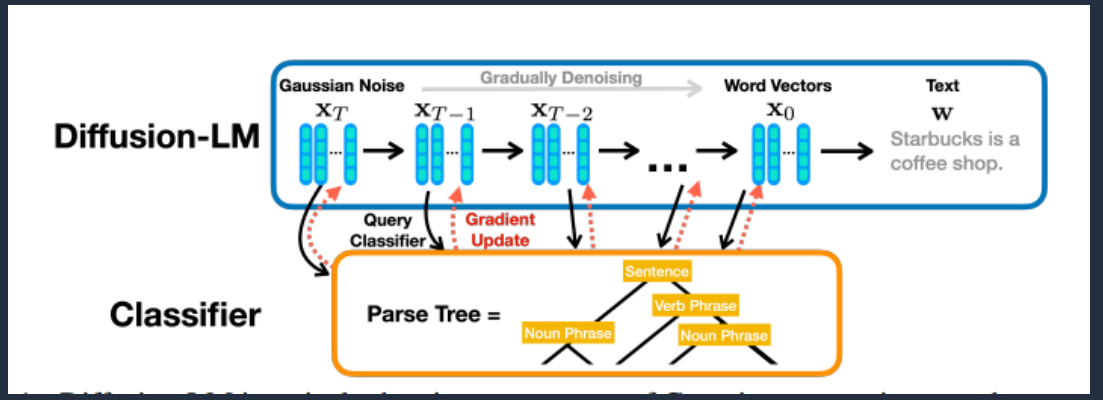
$$p(\mathrm{x}_{0:\mathrm{T}}|\mathrm{c}) = \prod_{t=1}^{T} p(x_{t-1}|x_t, c)$$

By Bayes rule,

$$p(x_{t-1}|x_t, c) \propto p(x_{t-1}|x_t)p(c|x_{t-1}, x_t)$$

Conditional independence

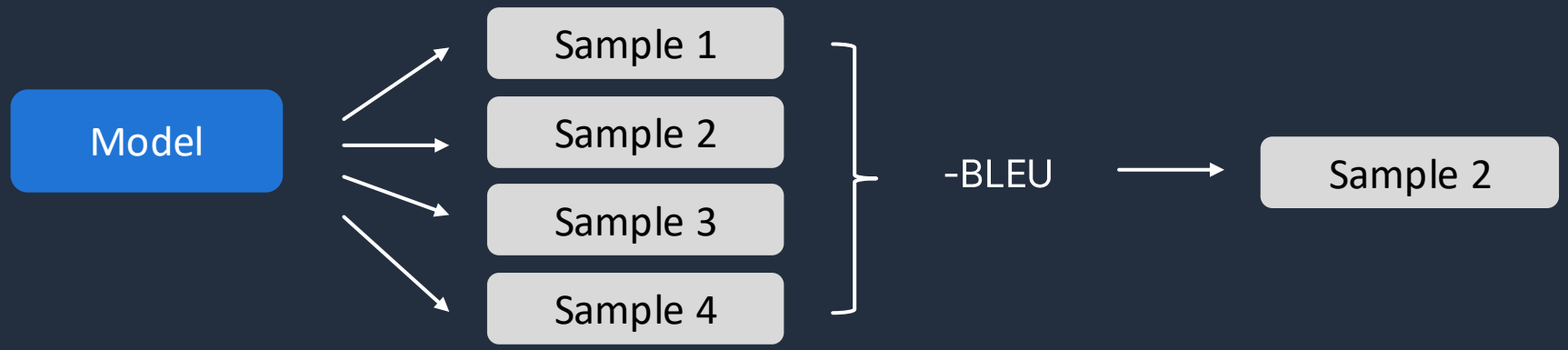
$$p(c|x_{t-1}, x_t) = p(c|x_{t-1})$$



$$\longrightarrow \quad \nabla \log p(x_{t-1}|x_t, c) = \nabla \log p(x_{t-1}|x_t) + \nabla \log p(c|x_{t-1})$$

Diffusion-LM

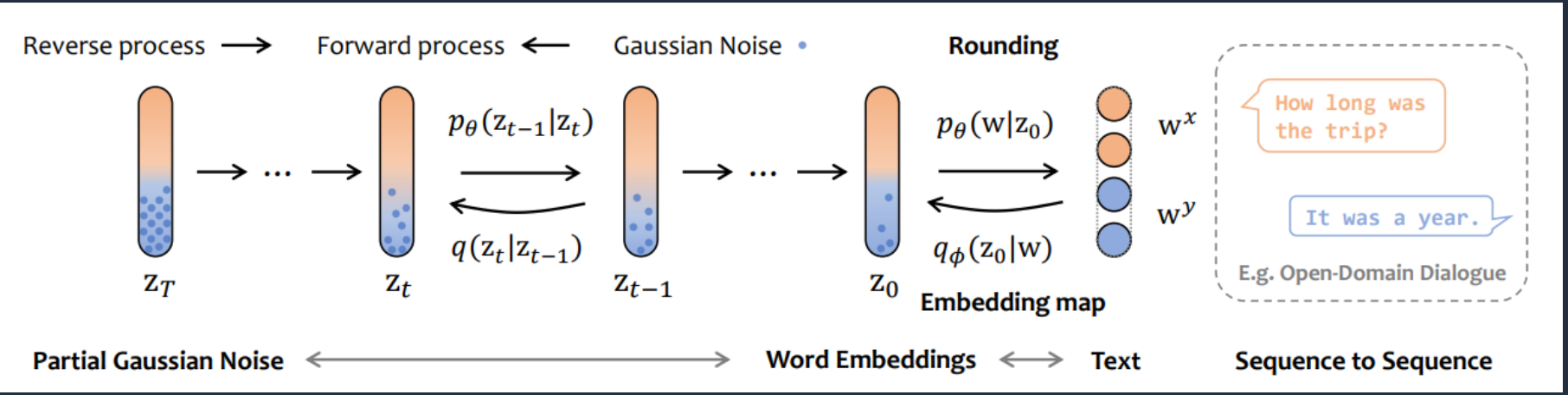
- MBR decoding
  - Diffusion generated samples might contain misaligned/non-fluent sentences
  -> Generate many candidates and then select the best



- One of the commonly used decoding methods to compensate for low-quality generation in DLMs

DiffuSeq

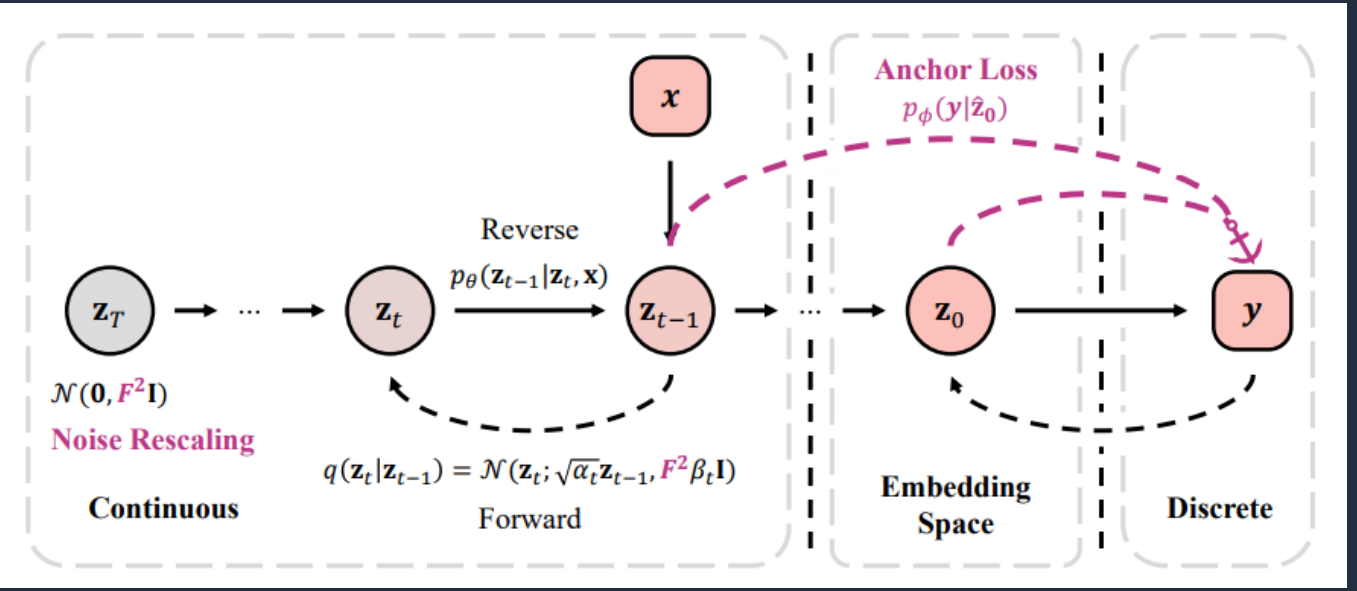
- Extension of Diffusion-LM → supports conditional (Seq2Seq) text generation.



- Partial noising: Add Gaussian noise only to target embeddings during forward process.
- Conditional denoising: Transformer denoiser recovers y_0 while attending to x_0 at every step.

Difformer

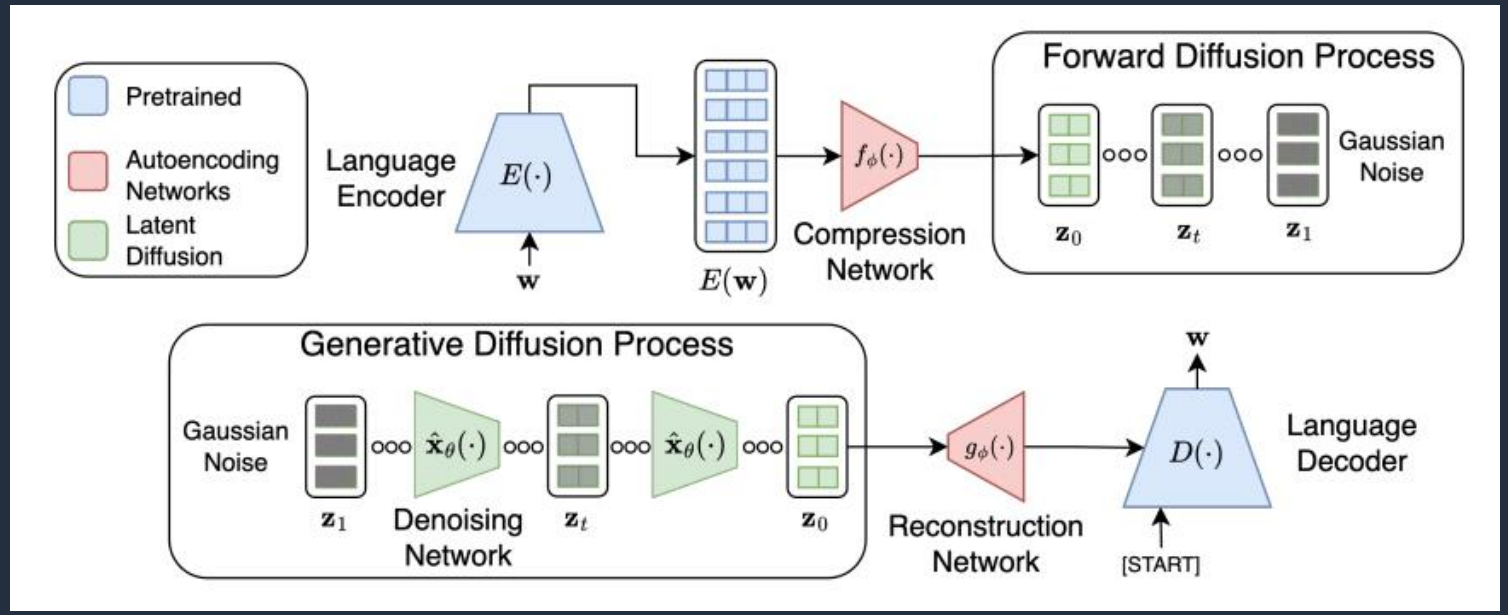
- **Embedding Collapse**: **Learnable embeddings** shift during training, leading to unstable training and reduced representation capacity
- **Denoising Model Degeneration**: Insufficient noise in **high-dimensional** embedding space



Gao et al., "Empowering Diffusion Models on the Embedding Space for Text Generation", NAACL 2024.

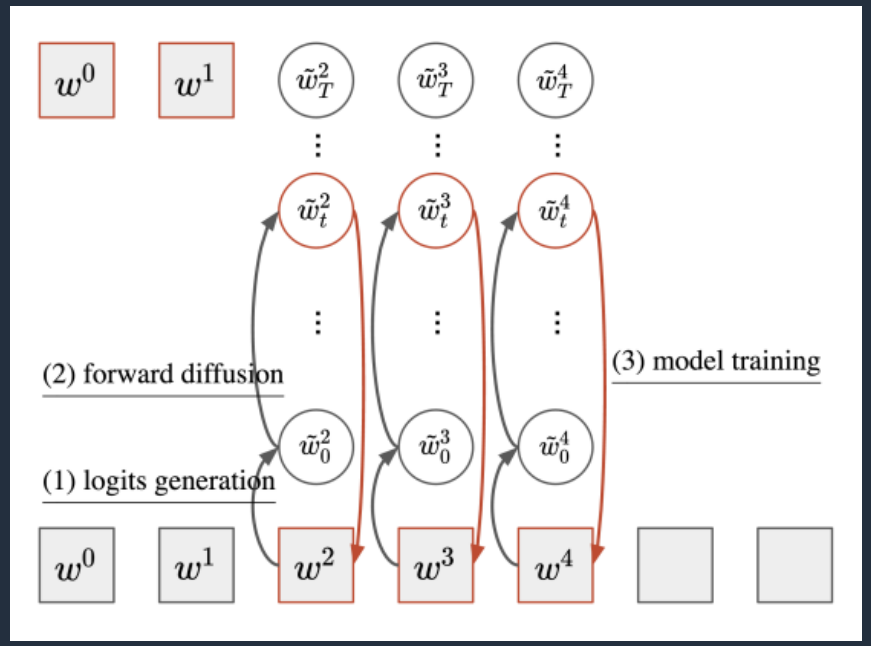
LD4LG

- Instead of utilizing trainable embedding, use encoder-decoder for latent diffusion
- Use a pretrained encoder–decoder LM (BART/T5) to build a language autoencoder.
- Compress encoder outputs into a fixed-length, continuous latent space → run diffusion here → reconstruct text via pretrained decoder.



SSD-LM

- Inherent limitation in diffusion LMs -> **Fixed length generation**
- **Instability** in embedding based formulation (+ classifier incompatibility)



- Propose semi-autoregressive decoding
  (parallel generation within each block, autoregressive generation across blocks)
- Operate directly in vocabulary space
  (no embedding, latent mapping)

$$\tilde{w}_{(i)} = \begin{cases} +K \text{ when } w = V_{(i)} \\ -K \text{ when } w \neq V_{(i)} \end{cases} \qquad \hat{w}_{(i)} = \begin{cases} +K \text{ if } i = \text{argmax}(\boldsymbol{w}_{\text{logits}}) \\ -K \text{ otherwise} \end{cases}$$

Han et al., "SSD-LM:Semi-Autoregressive Simplex-based Diffusion LM", ACL 2023.

Downstream Applications

- Easy to integrate attribute control (e.g., sentiment, style, topic) using external or learned classifiers.
- Stochastic, multi-step sampling naturally explores multiple plausible outputs.

⟶ Controllable generation, Textual style transfer, Artistic generation

Prefix: Horrible hotel. Horrible hotel. Serioulsy, horrible hotel.	**1-gram**	**2-gram**	**3-gram**
FT (Greedy decoding): The hotel was a nightmare. The rooms were filthy, the bathrooms were filthy, the food was disgusting, the food was disgusting. The food was disgusting. The food was disgusting. The food was disgusting. The food was disgusting. The food was disgusting....	The: 100.00%	The hotel: 100.00%	The hotel was: 100.00%
FT (Top-p sampling): Terrible, awful hotel. Horrible, awful hotel. I'm going to the awful hotel. I'll never get in. What the hell is this hotel? It's awful. I don't like it. Where? It's awful. I think they have a bathroom there. I'm going to the awful hotel. I'll never get in. I want out of here. This hotel is horrible. I'm going to the awful hotel. I'll never get in. I want out of here...	I: 13.39% Horrible: 12.20% The: 8.27% Terrible: 5.91% This: 4.33%	Horrible hotel: 12.50% Terrible hotel: 6.05% This is: 2.82% I hate: 1.61% I mean: 1.61%	Horrible hotel. Horrible: 3.48% Horrible hotel. I: 2.61% Terrible hotel. Terrible: 2.17%, Terrible hotel. Horrible: 1.30% I could not: 0.87%
Ours (Greedy decoding): Staff was unhelpful. So rude from the front desk. The hotel tried to lose the reservation of my family which I had with the hotel. This hotel is way overpriced. Hallways smelled bad. Carpet was disgusting. I'll never go back. I don't recommend the hotel. I can stay anywhere else!!	I: 10.55% The: 5.08% We: 3.52% Hotel: 2.73% They: 2.73%	I was: 1.56% Had to: 1.56% The room: 1.17% Dirty and: 1.17% I called: 1.17%	The room was: 1.18% The walls were: 0.78% I was told: 0.78% I called the: 0.78% I booked through: 0.78%

Zhang et al., "PLANNER: Generating Diversified Paragraph via Latent Language Diffusion Model", NeurIPS 2023.

Downstream Applications

Controllable Text Generation

Diffusion-LM (Li et al., 2022)

input (Semantic Content) output text	food : Japanese Browns Cambridge is good for Japanese food and also children friendly near The Sorrento .
input (Parts-of-speech) output text	PROPN AUX DET ADJ NOUN NOUN VERB ADP DET NOUN ADP DET NOUN PUNCT Zizzi is a local coffee shop located on the outskirts of the city .
input (Syntax Tree) output text	(TOP (S (NP (*) (*) (*)) (VP (*) (NP (NP (*) (*))))) The Twenty Two has great food
input (Syntax Spans) output text	(7, 10, VP) Wildwood pub serves multicultural dishes and is ranked 3 stars
input (Length) output text	14 Browns Cambridge offers Japanese food located near The Sorrento in the city centre .
input (left context) input (right context) output text	My dog loved tennis balls. My dog had stolen every one and put it under there. One day, I found all of my lost tennis balls underneath the bed.

SSD-LM (Han et al., 2022)

council will issue a decision in late September on whether to continue efforts to adopt the partnership model at the **[NO CONTROL]**

is one of the world's fastest-growing cities with over 4 million inhabitants. It is the most **[POSITIVE SENTIMENT]**

does not have the authority to regulate drug use on public property or punish people for it. The city **[NEGATIVE SENTIMENT]**

Li et al., "Diffusion-LM Improves Controllable Text Generation", NeurIPS, 2022.

Han et al., "SSD-LM:Semi-Autoregressive Simplex-based Diffusion LM", ACL 2023.

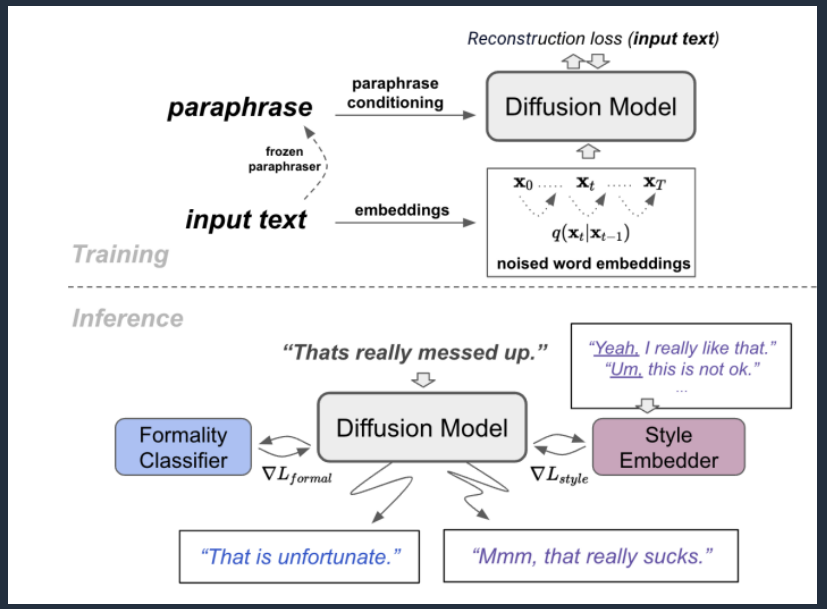
Downstream Applications

Text style transfer

Diffuseq-StylePTB
(Lyu et al., 2023)

Aspect	Transfers	Original Sentence	Additional Information	Transformed Sentence
Syntax	To Future Tense	She travels to Paris every summer to visit her family.		She **will** travel to Paris next summer to visit her family.
	To Present Tense	She had been studying architecture for five years.		She **has** been studying architecture for five years.
	To Past Tense	He walks to the store every day.		He **walked** to the store every day.
	Activate to Passive	The cat chased the mouse.		The mouse **was chased** by the cat.
	Passive to Activate	The proposal was approved by the committee yesterday.		The committee **approved** the proposal yesterday.
	PP Front to Back	Having watched the movie, they left the theater.		They left the theater after **having watched the movie**.
	PP Back to Front	They have been planning their vacation for months.		**For months**, they have been planning their vacation.

ParaGuide
(Horvitz et al., 2024)



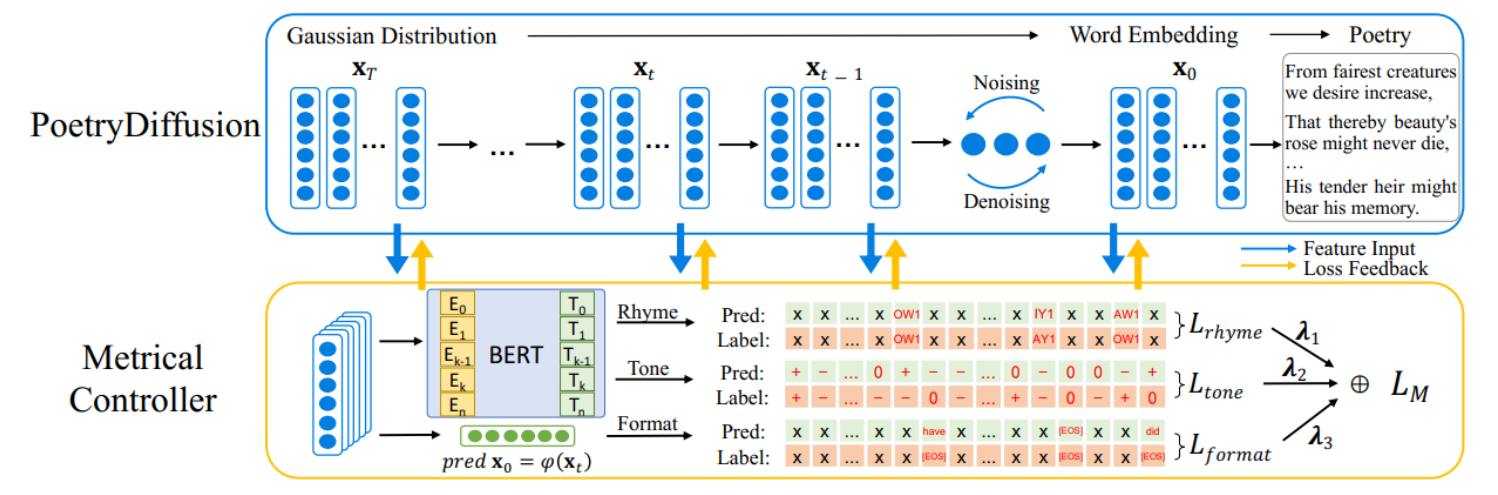
Lyu et al., "Fine-grained Text Style Transfer with Diffusion-Based Language Models", RepL4NLP 2023.

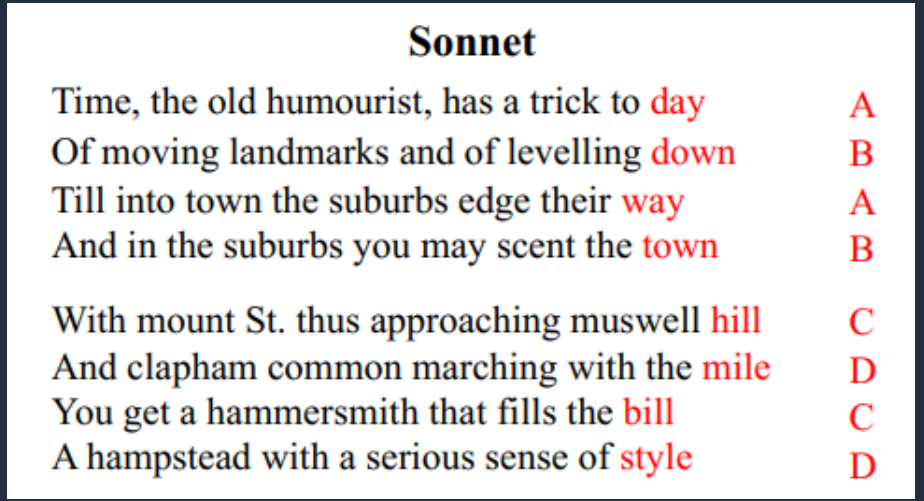
Horvitz et al., "ParaGuide: Guided Diffusion Paraphrasers for Plug-and-Play Textual Style Transfer", AAAI 2024.

Downstream Applications

Dialog/Poetry generation

PoetryDiffusion (Hu et al., 2024)





Sonnet

Time, the old humourist, has a trick to day	A
Of moving landmarks and of levelling down	B
Till into town the suburbs edge their way	A
And in the suburbs you may scent the town	B
With mount St. thus approaching muswell hill	C
And clapham common marching with the mile	D
You get a hammersmith that fills the bill	C
A hampstead with a serious sense of style	D

Hu et al., "PoetryDiffusion: Towards Joint Semantic and Metrical Manipulation in Poetry Generation", AAAI 2024.

Next week

Week 8
Diffusion Language Models II

- Discrete diffusion language models
- Masking & Absorbing state
- Token ordering

Week 8
Diffusion Language Models III

- AR VS. NAR
- AR-like formulation in DLMs
- Block-wise diffusion

Thank you!

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